



Anti Theft Locker System

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INTRODUCTION

Problem Overview

Traditional lockers rely on mechanical locks or basic PINs that are passive. They lack real-time monitoring and cannot protect assets once the primary defense is compromised by an intruder.

Project Importance

Integrating IoT with active defense (secret compartments) allows lockers to transform into smart security devices that provide real-time alerts and secondary failsafe mechanisms during theft attempts.

OBJECTIVES

AIM

Primary Project Aim

-  To develop a secure smart locker using ESP32 that detection intrusions via vibration sensors.
-  To implement a real-time IoT alert mechanism via Telegram/WhatsApp for immediate owner notification.
-  To automate a **hidden trapdoor mechanism** that physically secures valuables upon detecting unauthorized access.

METHODOLOGY

Working Principle

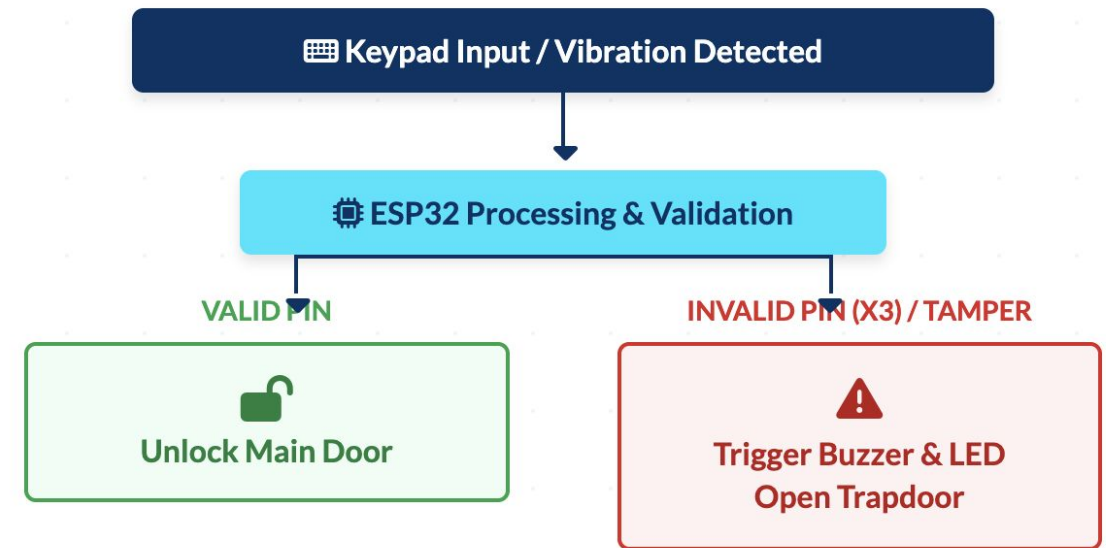
Input: User enters password via 4x4 Keypad.

Validation: ESP32 checks against stored PIN.

3-Strike Rule: 3 incorrect attempts trigger localized buzzer, LED, and remote alerts.

Intrusion: Vibration detection triggers the trapdoor servo, relocating assets.

Connectivity: Logs and status are synced to Firebase/React dashboard.



HARDWARE & SOFTWARE

Category	Component / Tool	Specifications
Microcontroller	ESP32-WROOM	Integrated WiFi + Bluetooth (BLE)
Sensor	SW-420 Vibration Sensor	Detects physical shocks and tampering
Actuators	SG90 Servo Motors (x2)	Main door control and Trapdoor relocation
Software IDE	Arduino IDE	Firmware development in C++
Web Stack	React & Firebase	Dashboard UI and Real-time database

RESULTS

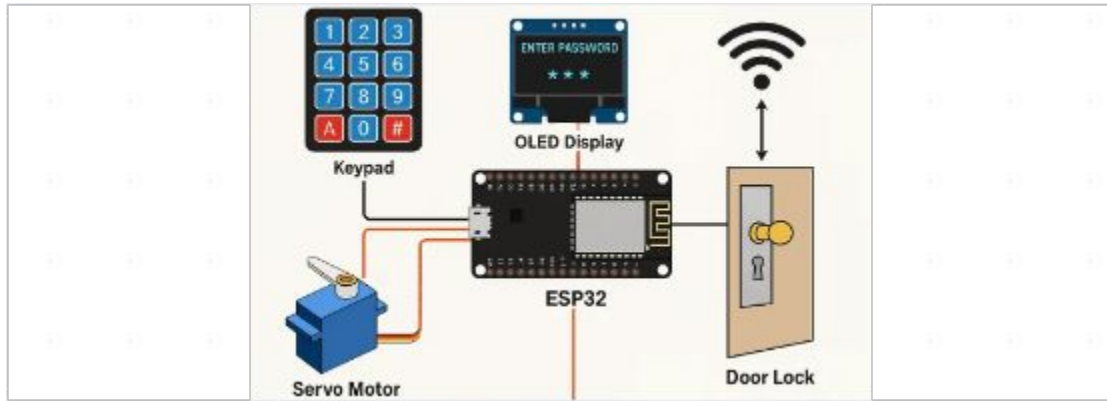


Fig 7.1: Prototype Circuit Assembly

System Performance

The system demonstrated a **response time of less than 1.5 seconds** for IoT alerts after the 3rd failed attempt. The vibration sensor effectively distinguished between ambient noise and intentional impacts during testing. Asset isolation via the trapdoor servo was successful in 100% of the simulated breach scenarios.

Conclusion

The Anti-Theft Smart Locker System successfully integrates **IoT** and embedded logic to provide an **active defense solution**. It offers a low-cost, reliable, and scalable approach to personal security that outperforms **traditional passive locking mechanisms**.

FUTURE WORK

-  **Biometric Authentication:** Integrating fingerprint or facial recognition for secondary verification.
-  **Power Resilience:** Implementing a high-capacity battery backup for uninterrupted operation during power failures.
-  **Visual Logging:** Using ESP32-CAM to capture and transmit images of the intruder.
-  **Fleet Management:** Scaling the dashboard to manage multiple lockers in bank or hostel environments.

REFERENCES



Books

Espressif Systems, *ESP32 Technical Reference Manual*, v4.1.



Websites

Arduino.cc - Documentation & Keypad Library Reference.
Firebase.google.com - Realtime Database Guides.



Research Papers

IEEE: "IoT Based Smart Security and Home Automation System" (2018).
IRJET: "Smart Door Lock System Using Arduino and IoT" (2020).

Thank You

Questions & Discussion

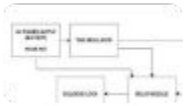


IMAGE SOURCES



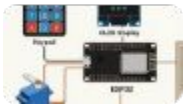
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